

ANALYSTLAB AFRICA
Experience Lab Internship Programme



HealthConnect Clinic
Model Testing & Refinement Report

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and Healthcare Support Using Data and AI
Track: Data Science

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Week 7 Submission – Model Testing, Error Analysis & Refinement

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This report is the Week 7 Data Science track deliverable for the HealthConnect Clinic Experience Lab engagement: the HealthConnect Model Testing & Refinement Notebook/Report. Where Week 6 focused on advancing and integrating the candidate model, Week 7 systematically tests that candidate rather than assuming its small reported improvement was robust. Testing is organised into eight tests, T01–T08, executed in the accompanying Week 7 notebook (Model_Testing_Error_Analysis_Refinement.ipynb) against the same patient-aware train/test split carried forward from Week 5, and every figure reported below is taken directly from that notebook's executed output.

In line with this track's established practice of treating executed code output as authoritative over narrative commentary, every metric in this report has been verified against the notebook cell that produced it. This is the second revision of the Week 7 testing notebook; one discrepancy flagged in the previous revision (a ROC-AUC/PR-AUC conflation in the final holdout interpretation) has since been corrected and is noted as resolved in Section 9. Two further issues remain open in this revision and are flagged explicitly, in Sections 7 and 14, rather than silently corrected or silently repeated.

1. Week 6 candidate model summary

The Week 6 deliverable produced a Refined Logistic Regression as the HealthConnect no-show candidate model, built with `class_weight="balanced"` on the same patient-aware Week 5 training split, and augmented with three engineered interaction features identified from the Week 6 error analysis: `history_x_lead_time` (`previous_no_shows` × `booking_lead_days`), `distance_x_lead_time` (`distance_to_clinic_km` × `booking_lead_days`), and `history_interaction` (`previous_appointments` × `previous_no_shows`), giving 38 modelling features in total. On the held-out test set, the candidate achieved a ROC-AUC of 0.6890 and a PR-AUC of 0.6717, a reported improvement of +0.0017 ROC-AUC and +0.0037 PR-AUC over the Week 5 baseline. A provisional operating threshold of 0.45 was selected ahead of the conventional 0.50 default, using training-only out-of-fold evidence, to favour no-show recall in line with HealthConnect's proactive-support objective.

The Week 6 report itself identified five limitations that this track carried into Week 7 as the testing agenda for this deliverable: the small reported improvement had not been tested for stability across validation samples; patient-grouped cross-validation could not be performed because `patient_id` is not present in the processed modelling matrices; segment-level performance had not yet been tested; the 0.45 threshold remained provisional; and the model's cross-track requirements had not been end-to-end validated. Each of these is addressed in Sections 2–13 below.

2. Testing approach

Testing follows an eight-test plan, summarised in Table 1, designed to answer the testing-readiness question for the Week 6 candidate: is the model's reported improvement reproducible, where does it make errors, does it behave consistently across HealthConnect patient and appointment segments, are its learned features stable, and does its provisional threshold represent a defensible operational trade-off.

Test ID	Test	Purpose
T01	Cross-validation stability	Assess whether candidate performance is consistent across validation folds, not a single lucky split.

Test ID	Test	Purpose
T02	Baseline vs candidate comparison	Determine whether the Week 6 improvement over the Week 5 baseline holds under a fair, fold-matched comparison.
T03	False-negative/false-positive analysis	Characterise the appointments the model misclassifies at the 0.45 threshold.
T04	Segment performance	Test whether error rates are concentrated in identifiable patient or appointment segments.
T05	Feature stability	Test whether the Week 6 engineered interaction features, and other predictors, remain stable across folds.
T06	Overfitting/generalisation	Compare training-fold and validation-fold performance for signs of overfitting.
T07	Operational threshold validation	Test whether the provisional 0.45 threshold produces a defensible recall/false-positive trade-off against the 0.50 default.
T08	Cross-track validation	Validate Data Analytics' Week 7 findings against the Data Science modelling sample (mandatory HC-POD cross-track activity).

All validation uses `StratifiedKFold(n_splits=5, shuffle=True, random_state=42)` on the Week 5 training partition (3,771 appointments). A patient-grouped cross-validation, which would test generalisation across individual patients rather than individual appointments, could not be reproduced in Week 7 because `patient_id` was not retained in the processed Week 5 training matrices; this is carried forward as an explicit validation-structure limitation (Section 14) rather than treated as resolved. Out-of-fold (OOF) predicted probabilities, generated with `cross_val_predict` so that every prediction comes from a fold that did not see that observation during training, are used throughout T03–T04 and T07 to keep the error and segment analysis leakage-safe.

A methodological note on the feature set used for this testing: the Week 7 notebook reconstructs the Week 6 candidate's engineered features before testing it, but its reconstruction function only recreates `history_x_lead_time` and `distance_x_lead_time`, reproducing 37 of the Week 6 candidate's 38 features and omitting `history_interaction`. This gap is addressed in Section 14 as a limitation of the Week 7 testing exercise itself; the figures reported in Sections 3–9 below reflect the 37-feature reconstruction actually executed, not the full 38-feature Week 6 specification.

3. Test results

3.1 T01 — Cross-validation stability

Table 2 reports the candidate's performance on each of the five stratified folds, at the provisional 0.45 threshold.

Fold	ROC-AUC	PR-AUC	Precision@0.45	Recall@0.45	F1@0.45
1	0.6666	0.6719	0.6230	0.6985	0.6586

Fold	ROC-AUC	PR-AUC	Precision@0.45	Recall@0.45	F1@0.45
2	0.6959	0.7038	0.6219	0.7165	0.6659
3	0.6898	0.6906	0.6267	0.7139	0.6675
4	0.6698	0.6683	0.6115	0.7423	0.6705
5	0.6519	0.6718	0.5996	0.7371	0.6613
Mean (std)	0.6748 (0.0179)	0.6813 (0.0153)	0.6165 (0.0110)	0.7216 (0.0179)	0.6647 (0.0048)

Performance is stable in direction — the candidate discriminates no-shows from attendances above chance in every fold — but the ROC-AUC range (0.6519 to 0.6959, a spread of 0.044) shows that the single-split ROC-AUC of 0.6890 reported in Week 6 sits inside normal fold-to-fold variation rather than representing a fixed, reproducible figure. T01 is treated as validated: the candidate is stable enough to proceed to the remaining tests, but its performance should be reported as a range, not a point estimate.

3.2 T07 — Operational threshold validation

The provisional 0.45 threshold was tested against the conventional 0.50 default using the same OOF probabilities (Table 3).

Threshold	Precision	Recall	F1	F2	False Positives	False Negatives	Flagged Rate
0.50	0.6382	0.6247	0.6314	0.6274	687	728	0.5036
0.45	0.6162	0.7216	0.6648	0.6978	872	540	0.6025

Lowering the threshold from 0.50 to 0.45 avoided 188 false negatives (missed no-shows) at the cost of 185 additional false positives (unnecessary interventions) — a trade-off of approximately 0.984 extra false positives for every false negative avoided, and an increase in the overall flagged rate from 50.4% to 60.3% of training appointments. T07 confirms that 0.45 meaningfully shifts the recall/false-positive balance, but the trade-off is close to one-for-one and remains provisional pending a HealthConnect-supplied tolerance for outreach capacity.

3.3 T06 — Generalisation/overfitting

Fold	Train ROC-AUC	Validation ROC-AUC	Generalisation Gap
1	0.6944	0.6666	0.0278
2	0.6843	0.6959	-0.0116
3	0.6877	0.6898	-0.0021
4	0.6926	0.6698	0.0228
5	0.6979	0.6519	0.0459
Mean	—	—	0.0166

The mean generalisation gap of 0.0166 is small and the per-fold gap is twice negative, so T06 finds no material overfitting: the candidate's training-fold and validation-fold ROC-AUC track closely enough that the Week 7 testing exercise is not undermined by the model simply having memorised the training data.

4. Model performance metrics

Table 4 reports the overall out-of-fold classification performance of the Week 6 candidate on the full Week 5 training partition (n = 3,771), at the provisional 0.45 threshold, using the 37-feature reconstruction described in Section 2.

Class	Precision	Recall	F1-score	Support
Attended (0)	0.64	0.52	0.58	1,831
No-Show (1)	0.62	0.72	0.66	1,940
Accuracy	—	—	0.63	3,771

The model correctly classifies 63% of appointments overall, with a recall of 0.72 for the No-Show class — the metric most directly tied to HealthConnect's objective of identifying patients likely to miss their appointment — but a lower recall of 0.52 for the Attended class, meaning a substantial share of patients who would have attended are nonetheless flagged as at-risk. This precision/recall profile is consistent with the T07 threshold analysis in Section 3.2: the 0.45 threshold deliberately trades Attended-class recall for No-Show-class recall.

5. Error analysis

At the 0.45 threshold, the OOF confusion matrix on the training partition is: 959 true negatives, 872 false positives, 540 false negatives, and 1,400 true positives, giving 2,359 correct predictions, 872 false positives and 540 false negatives out of 3,771 appointments (Table 5).

Outcome	Count
Correct	2,359
False Positive	872
False Negative	540

Table 6 compares the mean (standardised) feature values of correct predictions against each error type, for the five numerical predictors most relevant to HealthConnect's no-show hypothesis.

Error Type	Age	Booking Lead Days	Prev. Appointment s	Prev. No-Shows	Distance (km)
Correct	0.015	0.018	0.019	0.031	0.004
False Negative	0.037	-0.769	-0.087	-0.261	-0.203
False Positive	-0.063	0.428	0.001	0.078	0.115

False negatives — actual no-shows the model misses — are characterised by shorter booking lead times and fewer previous no-shows than correctly classified appointments; the model under-estimates no-show risk precisely for patients who look, on paper, like reliable attenders. False positives — attendees unnecessarily flagged — are characterised by longer booking lead times and greater distance from the clinic, indicating the model over-weights these two features as risk signals for some genuinely low-risk patients. Both patterns are consistent with, and are

investigated further by, the segment testing in Section 6 and the feature-stability testing in Section 8.

6. False-positive/false-negative analysis

T04 evaluated model performance across every categorical and binned segment available in the raw Week 5 training data — gender, age group, appointment type, appointment day, appointment time, reminder status, reminder channel, Sunday-appointment flag, booking-lead-time band and distance band — producing 40 segment/group combinations in total. Table 7 reports the segments with the highest false-negative rates (FNR, missed no-shows) and highest false-positive rates (FPR, unnecessary interventions) among groups with at least 50 observations, the practical screening threshold used in the notebook to avoid reporting unstable small-sample rates.

Segment	Group	N	FNR	FPR	Direction of concern
Booking lead time	0–7 days	480	0.876	0.051	High FNR — no-shows booked close to the appointment date are mostly missed.
Booking lead time	8–14 days	468	0.801	0.128	High FNR — same pattern, slightly attenuated.
Booking lead time	15–30 days	992	0.457	0.398	Elevated FNR.
Gender	Prefer not to say	79	0.378	0.381	Elevated FNR in a smaller demographic group.
Reminder channel	SMS	1,509	0.346	0.421	Elevated FNR.
Distance	20+ km	294	0.124	0.741	High FPR — distant patients are frequently over-flagged.
Booking lead time	31–60 days	1,831	0.057	0.910	Highest FPR — appointments booked far in advance are flagged almost indiscriminately.

The clearest pattern is a lead-time trade-off: short-lead-time appointments have very high false-negative rates (the model under-flags them) while long-lead-time appointments have very high false-positive rates (the model over-flags them), with the 31–60 day band in particular flagging 91.0% of attendees who did not no-show. Distance shows a similar, smaller effect, consistent with the error-analysis means in Section 5. A broader interpretation of every one of the 40 tested segments, including age group, appointment type, appointment time and reminder status, is included as supporting evidence in the notebook's own segment-performance tables; several of the notebook's own markdown commentary cells accompanying this test cite group labels (for example, an "appointment_type" group named "New Patient") that do not appear among the segment's actual four executed groups (Diagnostic Test, Follow-up, General Consultation, Specialist Consultation). This report relies only on the executed segment tables reproduced above and in the notebook's saved CSV outputs, not on that narrative text.

7. Baseline vs candidate/refined model comparison

T02 re-compared the Week 5 baseline Logistic Regression and the Week 6 candidate on the same five stratified folds, using matched fold indices so that both models are evaluated on identical validation rows (the baseline on its original 35-feature matrix, the candidate on the 37-feature reconstruction described in Section 2). Table 8 reports the fold-level ROC-AUC and PR-AUC for each model and the candidate-minus-baseline difference.

Fold	Baseline ROC-AUC	Candidate ROC-AUC	ROC-AUC Diff	PR-AUC Diff
1	0.6698	0.6666	-0.0032	-0.0057
2	0.7000	0.6959	-0.0041	-0.0009
3	0.6877	0.6898	+0.0021	+0.0005
4	0.6688	0.6698	+0.0010	+0.0035
5	0.6519	0.6519	+0.0001	-0.0004
Mean	—	—	-0.0008	-0.0006

On this fold-matched comparison, the candidate records a higher ROC-AUC than the baseline in 3 of the 5 folds and a higher PR-AUC in 2 of the 5 folds, but the mean difference across all five folds is slightly negative for both metrics (-0.0008 ROC-AUC, -0.0006 PR-AUC): the candidate wins more often, but by smaller margins than it loses by, so the net five-fold average is essentially flat to marginally unfavourable. This is a materially different, and more nuanced, picture than the single-split Week 6 holdout result (+0.0017 ROC-AUC, +0.0037 PR-AUC): under repeated validation, the Week 6 candidate's reported improvement over the baseline is not robust and should not be treated as a reliably reproducible gain.

A discrepancy identified during verification: the notebook's own narrative interpretation of this test, and its downstream Testing/Validation Record and Week 7 Project Summary cells, state that the candidate achieved a higher ROC-AUC in only 1 of 5 folds and a higher PR-AUC in 0 of 5 folds, with mean differences of +0.0001 ROC-AUC and -0.0004 PR-AUC. These figures do not match either the executed comparison table reproduced in Table 8 above or the notebook's own executed stream output for this test ("Candidate higher ROC-AUC in 3/5 folds", "Candidate higher PR-AUC in 2/5 folds", "Mean ROC-AUC: -0.0008", "Mean PR-AUC: -0.0006"), which this report has independently re-derived from the per-fold differences in Table 8 and confirmed to be internally consistent. In line with this track's practice of treating executed code output as authoritative over narrative commentary, this report uses the verified figures of 3/5, 2/5, -0.0008 and -0.0006 throughout, and flags the narrative's stale or erroneous 1/5, 0/5, +0.0001 and -0.0004 figures as an unresolved internal inconsistency in the Week 7 notebook (Section 14).

8. Refinement activities

T05 tested whether the Week 6 interaction features remained stable across folds by refitting the candidate five times and comparing coefficients. The two interaction features actually reconstructed in Week 7, `history_x_lead_time` (mean coefficient 0.0387, std 0.0152, positive in all five folds) and `distance_x_lead_time` (mean coefficient -0.0446, std 0.0216, negative in all five folds), were not identified as unstable. Two calendar-derived features, `appointment_day_of_year` (std 0.1856) and `appointment_month` (std 0.1564), showed considerably higher coefficient variability in magnitude, although their sign was consistent in

every fold (positive and negative respectively). Table 9 reports the full coefficient-stability comparison for these four features.

Feature	Mean	Std	Min	Max	Positive Folds	Negative Folds
appointment_day_of_year	0.4014	0.1856	0.1003	0.6692	5	0
appointment_month	-0.3581	0.1564	-0.5415	-0.0758	0	5
distance_x_lead_time	-0.0446	0.0216	-0.0785	-0.0130	0	5
history_x_lead_time	0.0387	0.0152	0.0248	0.0660	5	0

Since coefficient variability alone is not sufficient evidence to remove a feature, Week 7 tested a targeted calendar-feature ablation rather than deleting `appointment_day_of_year` and `appointment_month` outright: the candidate was refit without these two features on the same five folds and compared against the un-ablated candidate under a pre-declared performance tolerance of 0.002 (Table 10).

Metric	Mean (current)	Mean (ablated)	Mean change
ROC-AUC	0.6748	0.6753	+0.0005
PR-AUC	0.6813	0.6820	+0.0007
Recall@0.45	0.7216	0.7206	-0.0010
F1@0.45	0.6647	0.6636	-0.0011

All four mean changes fall within the ± 0.002 tolerance, with ROC-AUC and PR-AUC improving marginally and Recall@0.45 and F1@0.45 degrading marginally. Under the notebook's pre-declared decision rule (accept the simpler model only if ROC-AUC, PR-AUC and Recall@0.45 are not materially worse), the calendar-feature ablation was accepted: the Week 7 refined model drops `appointment_day_of_year` and `appointment_month`, reducing the feature count from 37 to 35, while retaining the two reconstructed Week 6 interaction features. The 0.45 operational threshold was retained unchanged, as no test in this report provided evidence to revise it beyond the trade-off already quantified in Section 3.2.

9. Retesting results

The ablated, 35-feature model was retrained on the full Week 5 training partition and evaluated once on the Week 5/6 held-out test set ($n = 615$ appointments), at the 0.45 threshold, for a final retest ahead of Week 8 (Table 11). As in Week 6, this holdout set has already been used for evaluation in earlier weeks and is therefore preserved for continuity rather than treated as a wholly unseen external validation set (Section 14).

Metric	Value
Accuracy	0.6315
Precision	0.6104
Recall	0.7267
F1	0.6635

Metric	Value
F2	0.7000
ROC-AUC	0.6876
PR-AUC	0.6719
True Negatives	259
False Positives	224
False Negatives	132
True Positives	351

The retested model's holdout ROC-AUC (0.6876) and PR-AUC (0.6719) are consistent with the cross-validated mean range reported in Section 3.1 (0.6748–0.6813), and its recall of 0.7267 at the 0.45 threshold confirms the same recall-favouring behaviour quantified in Table 3. The retest supports the Section 8 refinement decision: the simpler, ablated specification performs in line with expectations and introduces no new instability relative to the pre-ablation candidate.

A discrepancy flagged in the previous revision of this report has now been corrected in this updated notebook: its markdown interpretation of the final holdout evaluation previously conflated the holdout PR-AUC (0.6719) with the holdout ROC-AUC, which the executed `final_holdout_metrics` table in Table 11 above reports as 0.6876. The notebook's interpretation cell now correctly states both figures separately. The notebook's auto-generated README.md file, however, was not regenerated alongside this correction and still lists only "ROC-AUC: 0.6719" under its key metrics, so that file remains out of date relative to the notebook's own corrected interpretation and should be refreshed before Week 8.

10. Interpretation

Taken together, the eight Week 7 tests give a materially more cautious picture of the Week 6 candidate than the Week 6 report's single-split holdout result suggested. T01 shows the candidate is directionally stable but its ROC-AUC genuinely varies by over four points across folds; T02 shows its headline improvement over the Week 5 baseline does not survive a fair, fold-matched comparison, with a slightly negative mean difference across both discrimination metrics despite winning a majority of folds; T03 and T04 show that its errors are not random but concentrate in specific, identifiable groups — short-lead-time bookings for missed no-shows, long-lead-time and distant bookings for unnecessary flags; T05 and the Section 8 ablation show that two calendar features could be removed without materially changing performance, producing a simpler 35-feature specification; T06 rules out material overfitting as an explanation for any of the above; and T07 quantifies the 0.45 threshold's trade-off precisely, at close to one extra false positive for every false negative avoided.

The combined interpretation is that the Week 6 candidate is a genuine, if modest, improvement in form (a logistic regression with theoretically motivated interaction features and class-balancing) but not yet a demonstrated, robust improvement in validated out-of-sample performance over the Week 5 baseline. The Week 7 refinement (the calendar-feature ablation) simplifies the model without harming it, which is itself a useful, evidence-based outcome, but it does not resolve the underlying T02 finding. Both results should be reported to HealthConnect stakeholders together, rather than only the more favourable Week 6 figure.

11. Business relevance

HealthConnect's stated objective is to reduce missed appointments through proactive, risk-prioritised patient support, without imposing a disproportionate outreach burden on patients who would have attended regardless. The Week 7 testing results speak directly to that trade-off. The 0.45 threshold (Section 3.2) identifies roughly seven in ten actual no-shows (recall 0.7216–0.7267 across CV and holdout) at the cost of flagging around six in ten appointments overall for some form of intervention, with every additional no-show captured costing close to one additional unnecessary contact. The segment findings in Section 6 give HealthConnect a concrete, actionable refinement to that policy: because the model under-flags no-shows booked within 7–14 days of the appointment and over-flags appointments booked 31–60 days out, a uniform threshold applied across all lead times is not operationally efficient, and lead-time-aware intervention rules (for example, a lower risk threshold for short-lead-time bookings, or lighter-touch outreach for the 31–60 day band) would likely use outreach capacity more effectively than the single 0.45 cut-off currently in place.

At the same time, the T02 finding that the candidate's improvement over the simpler Week 5 baseline is not robust (Section 7) is directly relevant to any business decision to adopt the more complex, feature-engineered model in production: HealthConnect should not expect the Week 6 candidate to materially outperform the baseline in practice, and the business case for adopting it should rest on its interpretability and the stability of its interaction features (Section 8), not on an unreplicated discrimination gain.

12. Model suitability assessment

The Week 7 refined model (Logistic Regression, class-balanced, 35 features after the calendar-feature ablation, 0.45 provisional threshold) is assessed as suitable for continued integration testing as a decision-support, risk-prioritisation component, and is not assessed as suitable for production deployment as an autonomous or final decision-making system. This assessment rests on four considerations established in this report: moderate and fold-variable discrimination (Section 3.1); a confirmed, non-trivial precision/recall trade-off whose operational cost has not yet been formally accepted by a HealthConnect stakeholder (Sections 3.2 and 11); identifiable, non-random error concentration in specific booking-lead-time, distance and demographic segments (Section 6); and an unresolved weak-improvement finding relative to the simpler Week 5 baseline (Section 7). None of these considerations is disqualifying on its own, but together they mean the model's outputs should be used alongside clinical and operational judgement, within a bounded, monitored pilot, rather than as a standalone basis for intervention decisions.

The model's cross-track dependency with Data Analytics (Section 13) has passed, which removes one specific integration risk — that the lead-time and prior-history relationships embedded in the model's features might not match the independently validated Data Analytics findings — but two cross-track items (prediction-level appointment-ID traceability and a missing-distance data discrepancy) remain open and are carried into Section 14 as conditions on full Week 8 integration readiness.

13. Cross-track testing/integration evidence

The mandatory Week 7 HC-POD cross-track activity for this track is a Data Analytics → Data Science validation (T08), documented below in the Test → Finding → Action → Retest → Validated-outcome format required by the Week 7 assignment brief.

Field	Record
Track collaborated with	Data Analytics
Project dependency	Data Science's lead-time and prior-no-show-history features and modelling decisions should remain consistent with Data Analytics' independently validated findings and definitions.
Component/output tested	Data Analytics' Week 7 validated findings, tested against the Data Science modelling sample, out-of-fold predictions and segment-performance workflow.
Information/output received	Data Analytics Week 7 report: 4,737 eligible appointments; overall no-show rate 51.15%; 68.49% no-show rate for 41–60 day bookings; no-show rates by prior-no-show history of 46.30% (0), 55.87% (1) and 63.53% (2+); a combined 41–60 day/2+ prior no-show segment rate of 83.33% (130 of 156).
Information/output provided	Data Science's Week 7 OOF predictions, segment-performance results, model feature usage, and threshold/error analysis, plus the harmonised cross-track retest described below.
Testing activity completed	Compared the Analytics lead-time and history-group definitions and findings against the Data Science feature and segment-testing logic, then re-ran the Data Science segment test using the Analytics lead-time bands (0–7, 8–21, 22–40, 41–60 days) and history groups (0, 1, 2+ prior no-shows).
Issue/finding identified	The original Data Science lead-time segmentation used different cut-points from the Analytics report (Section 6 uses 0–7, 8–14, 15–30, 31–60, 61+ day bands). Exact no-show rates between the two tracks also cannot be expected to match, because Analytics reports the full eligible population (n = 4,737) while this notebook's segment testing uses only the Week 5 training subset (n = 3,771).
Refinement/action taken	Harmonised the Data Science retest to the exact Analytics bands and groups, and retained explicit population labels so that full-population Analytics rates are never presented as equivalent to model-sample rates.
Retest result	Using the harmonised bands, the Week 5 training subset shows no-show prevalence rising from 30.2% (0–7 days) to 39.7% (8–21 days) to 52.1% (22–40 days) to 68.2% (41–60 days), and from 46.6% (0 prior no-shows) to 55.5% (1) to 65.0% (2+). The combined 41–60 day/2+ prior no-show segment in the modelling sample (n = 129) shows a no-show rate of 84.5%, closely in line with the Analytics benchmark of 83.33% (n = 156) given the different populations. Both the lead-time direction and the prior-no-show-history direction were confirmed monotonically increasing, and the cross-track validation summary records PASS for both directional tests and for definition alignment.
What changed as a result	Data Science's segment-testing definitions are now explicit and aligned to the Data Analytics Week 7 bands, removing the risk

Field	Record
	that two tracks silently use different lead-time cut-points while describing the same underlying business relationship.
Evidence	Data Analytics Week 7 report; the notebook's cross_track_lead_results, cross_track_history_results and cross_track_validation_summary tables; and the saved CSV artefacts healthconnect_week7_cross_track_lead_time_retest.csv, healthconnect_week7_cross_track_history_retest.csv and healthconnect_week7_cross_track_validation_summary.csv.
How this improved the overall HealthConnect solution	Strengthens the evidential basis for the lead-time and prior-history predictors used in the Data Science model, reduces the risk of conflicting segment definitions being carried into Week 8 final integration, and gives both tracks a shared, reproducible segmentation to report from.

Two items raised in the Data Analytics Week 7 report are explicitly not claimed as resolved by this cross-track activity, and are carried forward to Section 14 and Section 15: prediction-level appointment-ID traceability (Analytics could not trace individual predictions back to a specific appointment_id in the evidence supplied), and a missing-distance data discrepancy (86 eligible records with missing distance data in the Analytics source versus 39 in the working dataframe used for this test).

14. Limitations and risks

14.1 Validation and methodology limitations

- Patient-grouped cross-validation is not possible in Week 7, because patient_id is not present in the processed Week 5 training matrices; StratifiedKfold on individual appointments is used as a practical alternative, which may understate the model's true generalisation risk across repeat patients.
- Preprocessing (scaling, encoding) was performed once, prior to the Week 7 fold split, rather than within a fully encapsulated per-fold pipeline; this introduces a small risk of optimistic performance estimates relative to a pipeline that fits preprocessing independently inside each fold.
- The final holdout set used for retesting (Section 9) has already been evaluated in earlier weeks and is therefore not a completely unseen, external validation set, reducing its evidential strength for a final go/no-go decision.
- The Week 7 notebook's own feature-reconstruction function only recreates two of the Week 6 candidate's three engineered interaction features (history_x_lead_time and distance_x_lead_time), omitting history_interaction (previous_appointments × previous_no_shows), with an unaddressed placeholder comment left in the notebook's own code for this third term. All Week 7 testing in this report, including the comparison against the Week 5 baseline and the final retest, was therefore conducted on a 37-feature reconstruction of the Week 6 candidate, not its full 38-feature specification, and this should be corrected before the model is treated as fully re-validated.
- The Week 7 notebook's own narrative interpretation of the T02 baseline-vs-candidate comparison (and the downstream Testing/Validation Record and Week 7 Project Summary cells that repeat it) states fold-win counts and mean differences that do not

match the test's own executed output, as set out in Section 7. This report has used the verified, internally consistent executed figures throughout; the inconsistency itself should be corrected in the notebook before it is carried into any Week 8 artefact.

- A related authoring gap flagged in the previous revision — the notebook's final-holdout interpretation conflating holdout ROC-AUC with holdout PR-AUC (both stated as 0.6719, against the executed value of 0.6876) — has now been corrected in the notebook's interpretation cell (Section 9). The auto-generated README.md was not regenerated alongside this fix and still states "ROC-AUC: 0.6719," so it should be refreshed before being treated as a reliable project record.

14.2 Model and operational risks

- The candidate's improvement over the Week 5 baseline is not robust under fold-matched comparison (Section 7); HealthConnect should not expect the added complexity of the engineered-feature model to deliver a reliably better discrimination score than the simpler baseline in production.
- The provisional 0.45 threshold increases no-show recall at a cost close to one additional unnecessary intervention for every missed no-show avoided, and no formal HealthConnect operational tolerance for this trade-off has yet been confirmed.
- Model performance is uneven across patient and appointment segments (Section 6), with short-lead-time bookings disproportionately under-flagged and long-lead-time and distant bookings disproportionately over-flagged; deploying a single uniform threshold without segment-aware adjustment risks both missed outreach and unnecessary intervention concentrated in specific, identifiable groups.
- The model's predictions are statistical associations, not causal explanations of why a given patient is likely to miss an appointment, and should not be used as the sole basis for an intervention or clinical decision.

14.3 Cross-track and integration dependencies

- Prediction-level appointment-ID traceability, raised in the Data Analytics Week 7 report, remains unresolved and should be addressed before Week 8 so that individual model predictions can be audited against source appointment records.
- The missing-distance discrepancy identified by Data Analytics (86 eligible records with missing distance data in the source versus 39 in the analyst working dataframe) remains open and has not been investigated by this track.
- ML Engineering pipeline validation — confirming that the saved model artefact, feature schema and threshold are consumed correctly by any downstream serving pipeline — has not yet been completed and is a further integration dependency for Week 8.

15. Week 8 recommendations

1. Correct the Week 7 feature-reconstruction gap identified in Section 14.1 by adding the missing `history_interaction` term, using the exact Week 6 formula, and re-run T01, T02 and the calendar-feature ablation on the complete 38-feature specification before the ablated model is finalised for Week 8.
2. Correct the T02 narrative/executed-output inconsistency identified in Section 7 in the notebook's own markdown cells and in the `testing_validation_record` and Week 7 Project Summary outputs, so that all Week 8 artefacts cite the verified figures (3/5 ROC-AUC

wins, 2/5 PR-AUC wins, mean differences of -0.0008 and -0.0006) rather than the unverified 1/5, 0/5, +0.0001 and -0.0004 figures.

3. Regenerate the notebook's auto-generated README.md so that its key-metrics section reflects the now-corrected holdout ROC-AUC of 0.6876 (Section 9), rather than the stale 0.6719 figure it still carries.
4. Freeze the final Week 7 feature specification, threshold and model artefact once the item above is corrected, and version them explicitly (feature schema, model file and threshold value) ahead of final integration.
5. Obtain a formal HealthConnect stakeholder decision on the acceptable operational trade-off between no-show recall and unnecessary-intervention burden (Sections 3.2 and 11), ideally informed by the lead-time-aware refinement suggested in Section 11, rather than carrying the 0.45 threshold into Week 8 as provisional by default.
6. Resolve the two outstanding Data Analytics cross-track items — prediction-level appointment-ID traceability and the missing-distance discrepancy (Section 13) — rather than treating the T08 cross-track activity as fully closed.
7. Complete ML Engineering pipeline validation of the final model artefact, feature schema and threshold, and resolve any preprocessing, feature-ordering or serialisation mismatch identified during that integration test.
8. Prepare concise Week 8 presentation evidence that follows the Week 6 output → Week 7 test → finding → refinement → retest → readiness-decision narrative established in this report, so the testing process itself, not only its final metrics, is visible to reviewers.

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